AJAY C

Embedded Firmware Engineer | B.E. Electronics & Communication Engineering | Tiruppur, Tamil Nadu

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SUMMARY

An industry-experienced fresher in real-time embedded systems, with hands-on experience in firmware development, debugging, and hardware validation. Applied system-level thinking to understand end-to-end functionality, analyze hardware-software interactions, and troubleshoot issues across system boundaries. Contributed to firmware development and testing for industrial gateway products, with a strong interest in continuous learning and professional growth in embedded systems.

INTERNSHIP EXPERIENCE

Embedded Intern – MachDatum(ThingConnect) , Coimbatore **2 Months | Paid Intern**

• Developed OTA firmware updates and user configuration features for a SIMCom A7672S-based Modbus RTU-to-GSM gateway.

• Implemented dual-bank flash management, firmware image verification([CRC-32 / SHA-256]), and bank switching for controlled OTA updates.

• Structured FreeRTOS tasks, priorities, and queues to maintain responsive Modbus polling during background OTA operations.

• Flashed firmware and performed functional testing on industrial Modbus-to-Wi-Fi and Modbus-to-Ethernet gateway PCBs.

• Managed firmware branches, commits, and collaborative development workflows using Git and GitHub.

TECHNICAL SKILLS

Languages: C (Embedded C), Python, basic C++ and Rust

Microcontrollers: STM32 (F411RE, F407, F030, G0C1, C5; ARM Cortex-M0+ and M4), ESP32, Arduino

Protocols & Networking: UART, I2C, SPI, CAN 2.0, Modbus RTU/TCP, Ethernet (TCP, HTTPS),MQTT

Peripherals: GPIO, external interrupts (EXTI), DMA, ADC, timers (PWM), RTC, watchdog

FreeRTOS: Native APIs, task scheduling, queues, semaphores, mutexes, software timers, ISR synchronization, stack and heap management

QNX Neutrino RTOS: Priority-based scheduling, message-passing IPC, real-time task management, process synchronization,

Development Tools: STM32CubeIDE, Keil, VS Code, Git, GitHub , QNX Momentics IDE

DEBUGGING & HARDWARE ANALYSIS

Tools & References: ST-LINK debugger, logic analyzer (PulseView), multimeter, datasheets, reference manuals and errata sheets

• **Firmware Debugging:** Debugged STM32 firmware using ST-LINK, breakpoints, step execution, register and memory inspection, and SWV trace output.

• **RTOS Analysis:** Traced FreeRTOS task transitions, context switches, synchronization states, stack usage, and heap allocation.

• **Protocol Analysis:** Captured and analyzed communication traffic using a logic analyzer to verify data flow, frame structure, and bus-level behavior.

• **Hardware Investigation** Identified a mismatch between STM32G0B0RET6 documentation and actual hardware support for bank swapping, confirmed through the official ST Community, and adapted the OTA bootloader design.

PROJECTS

5-DOF Robotic Arm | *STM32F411RE, ESP32*

- Built a robotic arm controlled in real time from a mobile app over Wi-Fi.
- Control path: Mobile app → Wi-Fi → ESP32 → UART → STM32F411RE → PWM → 5 servo motors.
- Refined servo actuation precision and reduced command latency between the app and the arm.

Distributed ECU Network with CAN, FreeRTOS & UDS Diagnostics

• Developed a 2-ECU automotive network using STM32F407, FreeRTOS, CAN(MCP2551) and UDS for real-time communication and diagnostics.

• Implemented sensor interfacing, fault detection, DTC management, LCD monitoring, and CANbased data exchange between ECUs.

• Designed a UDS diagnostic ECU to read/clear DTCs and log diagnostic events to an SD card with RTC timestamps.

Autonomous Vehicle Safety Supervisor | QNX Neutrino (SDP 8.0), C, Raspberry Pi 5 · QNX eHACK 2026

• Architected a 10-process QNX system where a high-priority supervisor checks navigation commands against braking distance and sensor data before allowing, limiting, or rejecting them.

• Implemented fault detection and recovery using watchdogs, heartbeat monitoring, sensor validation, 2-of-3 sensor voting, and automatic process restart.

• Ensured predictable real-time safety responses using priority scheduling, deadlock-free IPC, and memory locking

Adaptive Software-Defined Sonar Transmitter | STM32F407

• Designed an embedded control architecture to adapt sonar transmission parameters to changing underwater conditions.

• Generated LFM chirp waveforms using timer-triggered DAC output and DMA-based sample transfer.

• Implemented rule-based firmware to adjust transmission frequency, bandwidth, and pulse duration based on environmental inputs.

EDUCATION

B.E. Electronics & Communication Engineering *2024 – 2028*

KPR Institute of Engineering & Technology, Coimbatore, Tamil Nadu

CGPA: 7.6/10 | Class XII: 84.16% | Class X: 75.8%

CERTIFICATIONS & COURSES

- ARM Cortex-M3 & M4 Programming — Udemy
- Industrial, IoT & Consumer Microcontrollers — Infineon Technologies
- Embedded Systems for Beginners — NIELIT Calicut
- Electronics Foundation Course — Etalvis Learning
- C & Python Programming — Udemy
- PALS Innowah Cluster Finals (IIT Alumni) — Competition Certificate